

example: LTI System

$$y[n] - \frac{1}{9} y[n-2] = x[n]$$

2') Freq. Response?

$$H(e^{j\omega}) = \frac{1}{1 - \frac{1}{9} e^{-j2\omega}} = \frac{1}{(1 - \frac{1}{3} e^{-j\omega})(1 + \frac{1}{3} e^{-j\omega})}$$

2'2') $h[n] = ?$

$$1 \rightarrow (e^{j\omega}) = \frac{\cancel{A}^{1/2}}{1 - \frac{1}{3} e^{-j\omega}} + \frac{\cancel{B}^{1/2}}{1 + \frac{1}{3} e^{-j\omega}}$$

$$z = e^{-j\omega}$$

$$A = \left(1 - \frac{1}{3}S\right) H(S) \Big|_{S=3} = \frac{1}{1 + \frac{1}{3}S} \Big|_{S=3} = \frac{1}{2}$$

$$B = \frac{1}{2}$$

$$h(n) = \frac{1}{2} \left(\frac{1}{3}\right)^n u(n) + \frac{1}{2} \left(-\frac{1}{3}\right)^n u(n)$$

iii) $y[n]$? when $x[n] = \left(\frac{1}{3}\right)^n u[n]$

$$x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega}) = \frac{1}{1 - \frac{1}{3} e^{-j\omega}}$$

$$Y(e^{j\omega}) = X(e^{j\omega}) \cdot H(e^{j\omega}) = \frac{1}{(1 - \frac{1}{3}e^{-j\omega})^2 (1 + \frac{1}{3}e^{-j\omega})}$$

$$Y(e^{j\omega}) = \frac{\cancel{A}^{1/4}}{1 + \frac{1}{3} e^{-j\omega}} \cdot \frac{\cancel{B}^{1/4}}{1 - \frac{1}{3} e^{-j\omega}} \cdot \frac{\cancel{C}^{1/2}}{(1 - \frac{1}{3} e^{-j\omega})^2}$$

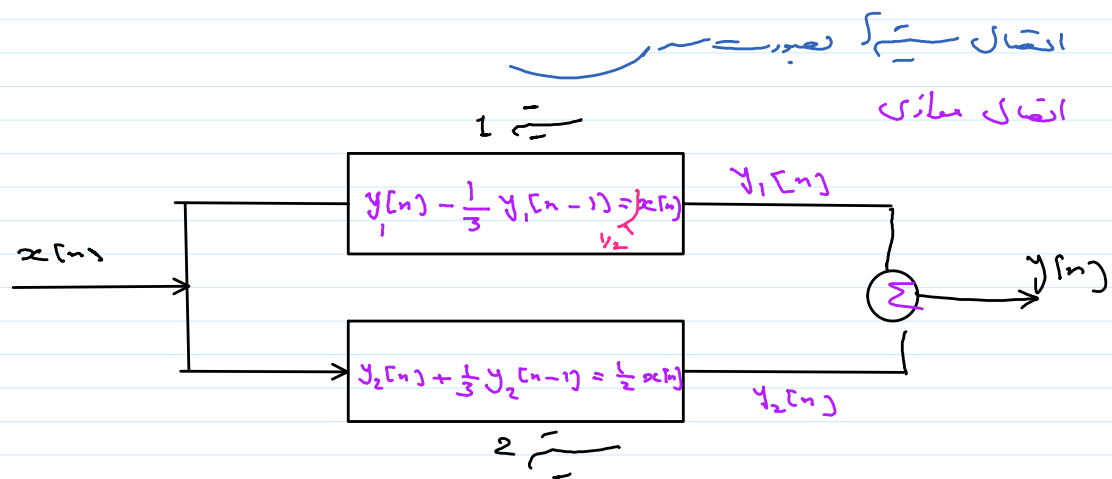
$$B = \frac{d}{ds} \left[\left(1 - \frac{1}{3}s\right)^2 \gamma(s) \right]_{s=3} = 1/4$$

$$y(n) = \frac{1}{4} \left(-\frac{1}{3}\right)^n u(n) + \frac{1}{4} \left(\frac{1}{3}\right)^n u(n) + \frac{1}{2} (n+1) \left(\frac{1}{3}\right)^n u(n)$$

$$H(e^{j\omega}) = \frac{1}{(1 - \frac{1}{3}e^{-j\omega})(1 + \frac{1}{3}e^{-j\omega})} = H_1(e^{j\omega}) \cdot H_2(e^{j\omega})$$

$$= \frac{1/2}{1 - \frac{1}{2} e^{-j\omega}} + \frac{1/2}{1 + \frac{1}{2} e^{-j\omega}} = H_1(e^{j\omega}) + H_2(e^{j\omega})$$

انفال سے کہ تصور ہے



$x[n] \equiv \text{input}$

$y[n] \equiv \text{output}$

input-output

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

$$y[n] + \frac{1}{4} y[n-1] + w[n] + \frac{1}{2} w[n-1] = \frac{2}{3} x[n] \quad (I)$$

$$y[n] - \frac{5}{4} y[n-1] + 2w[n] - 2w[n-1] = -\frac{5}{3} x[n] \quad (II)$$

DTFT

$$(I) \quad (1 + \frac{1}{4} e^{-j\omega}) Y(e^{j\omega}) + (1 + \frac{1}{2} e^{-j\omega}) W(e^{j\omega}) = \frac{2}{3} X(e^{j\omega})$$

$$(II) \quad (1 - \frac{5}{4} e^{-j\omega}) Y(e^{j\omega}) + (2 - 2 e^{-j\omega}) W(e^{j\omega}) = \frac{5}{3} X(e^{j\omega})$$

Eliminate $W(e^{j\omega})$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{3 - \frac{1}{2} e^{-j\omega}}{1 - \frac{3}{4} e^{-j\omega} + \frac{1}{8} e^{-j2\omega}}$$

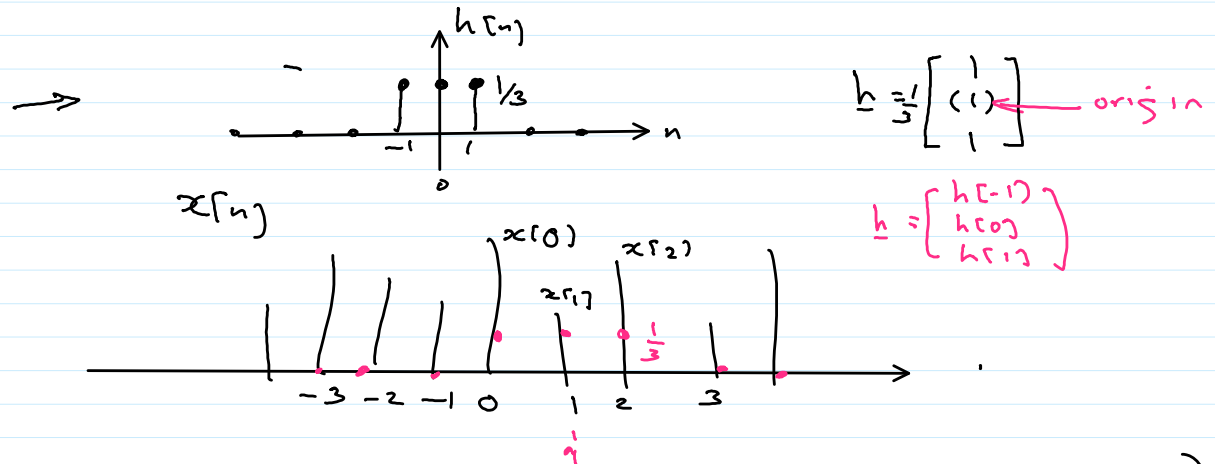
$$y[n] - \frac{3}{4} y[n-1] + \frac{1}{8} y[n-2] = 3x[n] - \frac{1}{2} x[n-1]$$

$$H(e^{j\omega}) = \frac{3 - \frac{1}{2} e^{-j\omega}}{(1 - \frac{1}{2} e^{-j\omega})(1 - \frac{1}{4} e^{-j\omega})}$$

$$= \frac{A}{1 - \frac{1}{2} e^{-j\omega}} - \frac{B}{1 - \frac{1}{4} e^{-j\omega}}$$

$$h[n] = 4 \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{4}\right)^n u[n]$$

← FIR ≡ Finite Impulse Response
IIR ≡ infinite I.R. →



$$x[-1] \cdot \frac{1}{3} + x[0] \cdot \frac{1}{3} + x[1] \cdot \frac{1}{3} = \frac{1}{3} (x[-1] + x[0] + x[1])$$

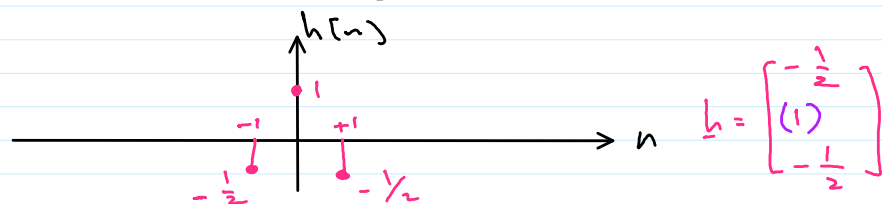
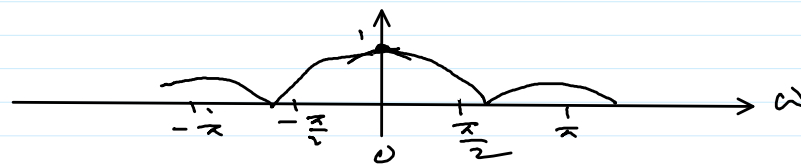
$$\sum_{k=-\infty}^{+\infty} x[k] h[n-k]$$

$$\text{DTFT}\{h[n]\} = H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n] e^{-jn\omega}$$

$$= \sum_{n=-1}^{+1} \left(\frac{1}{3}\right) e^{-jn\omega} = \frac{1}{3} (e^{+j\omega} + 1 + e^{-j\omega})$$

$$= \frac{1}{3} (2 \cos(\omega) + 1)$$

$$1.1 = \frac{2}{3} \cos(\omega) + \frac{1}{3}$$



$$y[n] = \sum_{k=-M}^{+M} a_k x[n-k]$$

$$y[n] - a y[n-1] = x[n]$$

$$H(e^{j\omega}) = \frac{1}{1 - a e^{-j\omega}}$$

$$h[n] = a^n u[n]$$

$$H(e^{j\omega}) = \frac{1}{1 - a e^{-j\omega}}$$

$$h[n] = a^n u[n]$$

$$a = \frac{1}{2}$$